

Sentio Network: The Decentralized Data and Compute Layer

[1 Introduction](#)

[1.1 Developer-First & Modular](#)

[2.3 Trustless & Incentive-Aligned](#)

[2.4 AI-Native](#)

[3 Design Overview](#)

[3.1 The Processor](#)

[3.2 Sentio Node](#)

[3.2.1 Data Storage](#)

[3.2.2 Processor Management](#)

[3.2.4 Staking and voting](#)

[4 Network Participation](#)

[4.1 Developers & Analysts](#)

[4.2 Sentio Node](#)

[4.3 Stakers / Delegators](#)

[5. Token Economy](#)

[5.1 Sentio Token Utility](#)

[5.2 Sentio Units \(SU\)](#)

[5.3 Job Assignment](#)

[5.4 Staking Model](#)

[5.5 Incentives](#)

[6. Summary](#)

1 Introduction

Modern Web3 applications generate vast volumes of on-chain and off-chain data across multiple chains and execution environments. Historically, this data has been scattered across siloed tools and custom pipelines, creating a fragmented data stack that increases integration overhead and slows development.

Sentio solves this fragmentation with a unified Web3 observability and data platform. It brings indexing, querying, visualization and alerts together into a single, coherent system, enabling developers to reason about smart contracts and application behavior more efficiently. Today, Sentio is widely adopted by leading protocols and development teams as production-grade infrastructure.

As the scale and importance of Web3 applications grow, so do expectations around data infrastructure. Developers increasingly seek not only powerful tooling, but also transparency, resilience, and shared participation in the systems that process and store critical application data.

The Sento Network extends the Sento platform by moving its **storage and processor layers into a decentralized network**. Instead of being operated by a single service provider, data storage and processor execution are handled by a distributed set of network participants. By decentralizing these core layers, the Sento Network enables verifiable data pipelines and aligned incentives for node operators, developers, and end-users—while preserving the developer experience and tooling that have made Sento widely adopted.

Sento Network is guided by three core principles that define both the product vision and the network design:

1.1 Developer-First & Modular

Sento is purpose-built for builders. Processors, dashboards, and alerts are designed to be intuitive, composable, and fast—so teams can go from idea to production without wrestling with low-level blockchain plumbing.

The network features a fully modular architecture with multiple layers. These layers are designed to function both in close coordination and as standalone components.

2.3 Trustless & Incentive-Aligned

Sento eliminates reliance on centralized providers by distributing indexing and governance across independent node operators. Powered by the \$ST token, economics are transparent and usage-driven. This ensures reliability, resilience, and sustainable growth of the ecosystem.

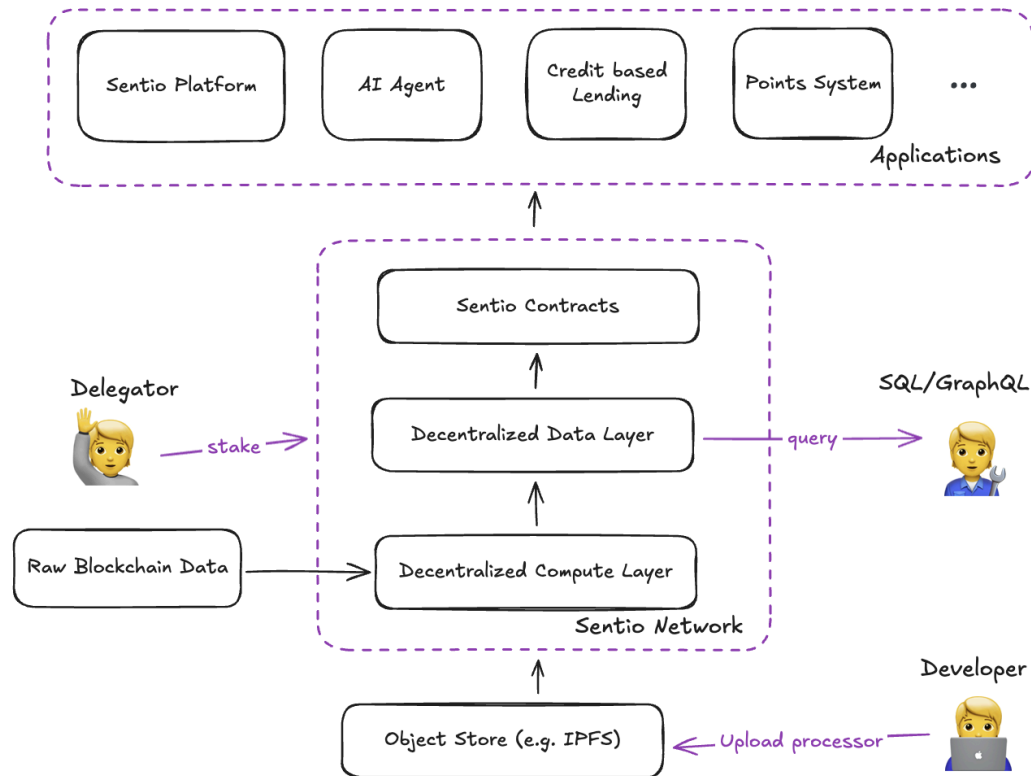
2.4 AI-Native

The entire Sento ecosystem is powered by AI. Natural-language queries, automated debugging, and predictive analytics are integrated into the workflow. Through the MCP (Model Context Protocol) service layer, developers can also build custom AI assistants that seamlessly consume blockchain observability data and extend into agent-based application

3 Design Overview

The core purpose of Sento's data collection is to transform the vast and raw blockchain data (like transactions and events) into structured, meaningful, and actionable insights. Instead of dealing directly with low-level blockchain details, Sento allows you to focus on the data that is relevant for your contracts.

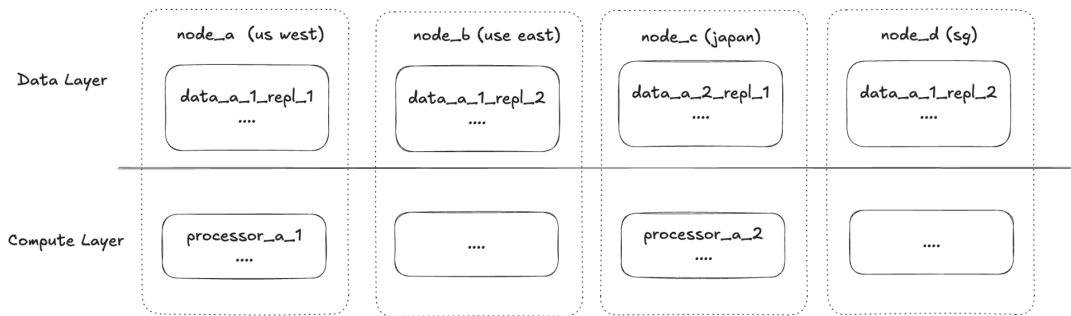
The **Sentio Network** is the decentralized infrastructure that facilitates this process. The broader **Sentio ecosystem** consists of three main components: the processor, the Sentio Network, and various applications.



- **Processor:** Defines custom logic for data indexing and ingestion.
- **Sentio Network:** decentralized network of **Sentio Nodes** organized into three independent layers:
 - **Compute Layer:** Executes processors for real-time computation, aggregation, analytics, and supports streaming, alerting, and visualization in coordination with the Data Layer.
 - **Data Layer:** Stores blockchain and processor-indexed data, accessible via APIs or SQL-like queries.
 - **Sentio Contract:** store metadata of data layer and compute layer, provide staking and voting mechanism.

This compute layer and data layer can scale independently. For example, a processor might have two replicas running on different continents. However, data generated by each processor can also be replicated to different nodes in nearby locations, even if

those nodes are not running that specific processor. This design offers maximum flexibility in terms of availability and cost.



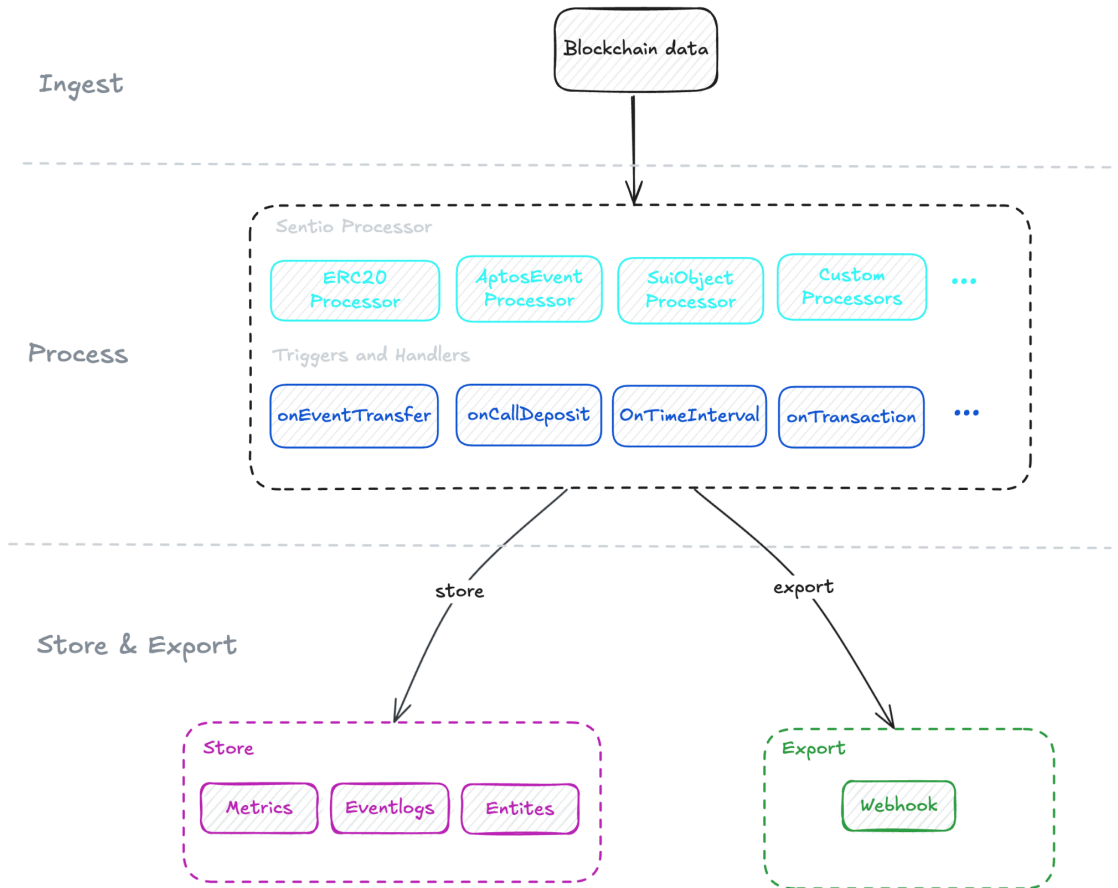
- **Applications:** Built on top of the Sentio Network, multiple applications will be developed, including but not limited to:
 - **Sentio Platform:** Provide intuitive UI and high-level functions to access and utilize the Sentio Network. The current centralized Sentio service will be migrated to the open platform.
 - **AI Applications:** Sentio indexed data can be directly consumed by AI models or agents as a high-quality, structured data layer, enabling intelligent analysis, automation, and decision-making.
 - **Decentralized Apps:** Applications that utilize Sentio indexed data, e.g. a credit-based lending protocol that considers lenders' historical debt repayment behavior, or a transparent points program that community could track and verify the points campaigns.

3.1 The Processor

At the heart of Sentio is the Processor. A Processor is typically a TypeScript program (but also supports multi-language) that you write using the Sentio SDK. It contains the logic that defines:

1. Processor Type (What to watch): Which smart contracts, wallets, or specific blockchain primitives (e.g. Object) to monitor.
2. Triggers and Handlers (How to react): What to perform when specific events happen.
3. Data Type (What to output): How to record or export the data

For backward compatibility and easy migration, the processor layer also supports other indexing syntax such as Subgraph, even though we encourage using Sentio SDK for its better program model and efficiency.



3.2 Sento Node

A Sento Node is the backbone of the Sento network. Each node combines three critical roles: managing processors, storing data, and participating in consensus.

3.2.1 Data Storage

Consists the Data Layer of the network.

- Persists data in the local Data store, and also provide interface for remote data join
- Exposes APIs (SQL, GraphQL, Analytic) for consumption.

3.2.2 Processor Management

Consists the Compute Layer of the network. The main job of a Sento node is to manage and run the processor.

- **Processor Scheduling**
 - Accepts assignments from contract.
 - Handles retries, backoff, and synchronization when nodes fall behind.
- **Processor Runtime**

- Runs processors in a containerized, sandboxed environment.
- Sentio-native processors communicate with the node via a gRPC-based driver.
- Subgraph processors are compiled to and executed in WebAssembly (WASM).
- **Billing & Receipts**
 - Generates signed receipts for indexing, queries, and RPC calls.
 - Aggregates receipts for periodic token settlement.
- **Security**
 - Fraudulent assignments or receipts can be challenged during dispute windows.
 - Proven misbehavior leads to slashing of staked tokens and loss of indexer reputation.
 - Future upgrades include proof-of-index for stronger correctness guarantees.

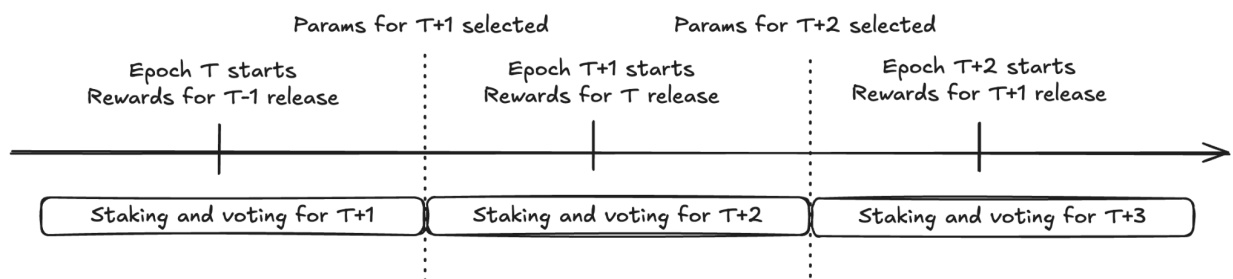
3.2.4 Staking and voting

Indexers are required to stake tokens in order to participate in the network and take part in governance voting.

Certain protocol parameters are governed through an epoch-based voting mechanism. Examples of adjustable parameters include, but are not limited to, the epoch length, the baseline SU exchange rate, and slashing ratios under different conditions. During each voting period, participating nodes submit their proposed parameter values to the Controller smart contract. Once the voting period concludes, the final parameter values are computed on-chain according to predefined aggregation rules.

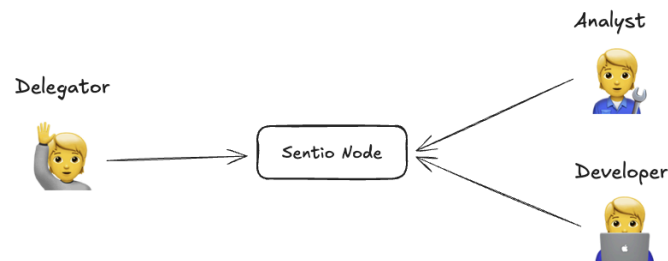
In addition to parameter voting, the protocol supports a proposal-based governance flow. Governance proposals are published to smart contracts and may be executed automatically once they are approved by a majority of staked voting power. This mechanism is intended for more complex governance actions, such as upgrading protocol contracts to new implementations.

The Sentio Network operates in discrete epochs. Any changes to stake—whether by indexers or delegators—only take effect at the beginning of the next epoch and do not apply immediately.



4 Network Participation

Sentio's decentralized observability network is powered by multiple stakeholders. Each role has distinct responsibilities and incentives, ensuring a balanced and sustainable ecosystem.



4.1 Developers & Analysts

- **What they do:**
 - Upload processors (analytics workflows) to the network.
 - Consume indexing, query, and RPC services.
- **Why they participate:**
 - Access transparent, decentralized analytics infrastructure.
 - Avoid vendor lock-in and opaque SaaS pricing.

4.2 Sentio Node

- **What they do:**
 - Run processors, store data in DB, and serve queries.
 - Provide access to Sentio Contract.
 - Compete to provide reliable and low-latency services.
- **Why they participate:**
 - Earn SU for every indexing and query job served.
 - Receive delegated stake from token holders to scale operations.
Gain reputation, leading to more workload assignments.

4.3 Stakers / Delegators

- **What they do:**
 - Stake \$ST tokens directly or delegate to Indexers.
 - Share in revenue generated by the nodes they support.
- **Why they participate:**
 - Earn passive income via revenue sharing.
 - Contribute to network security and decentralization.
 - Support trusted Indexers without running infrastructure themselves.

5. Token Economy

5.1 Sentio Token Utility

The **Sentio Token (\$ST)** serves as the native utility and governance asset of the protocol:

1. **Payment:** To access blockchain data, users consume Sentio Units (SU), settled via the native peer-to-peer unit of account, \$ST. Payments are distributed at the protocol level to network participants, including node operators and stakers. A small portion of each payment may be burned as part of the network's economic balancing mechanism.
2. **Staking:** As a guarantee of service standards, Sentio Nodes stake \$ST tokens in order to qualify to participate in securing the network. \$ST token holders may also delegate their stake to node operators, and in doing so, agree operator-defined commission to the relevant node operator for operating and maintaining indexing and computation services. Conversely, **\$ST** is used as a deterrent for penalising nodes or service providers for misbehaviour, with penalties including **slashing (i.e. deducting and burning the stake)**.
3. **Governance:** To promote decentralised community governance for the network, \$ST token holders would be able to vote on protocol-level parameters and upgrades. The right to vote is restricted solely to voting on features of Sentio; it does not entitle \$ST holders to vote on the operation and management of the Company, its affiliates, or their assets or the disposition of such assets to token holders, or select the board of directors or similar bodies of these entities, or determine the development direction of these entities, nor does \$ST constitute any equity interest in any of these entities or any collective investment scheme; the arrangement is not intended to be any form of joint venture or partnership. After governance launch there will be no individual or corporate entity or other active promoter, sponsor, or group or affiliated party that maintains sole control over Sentio.

5.2 Sentio Units (SU)

Sentio introduces Sentio Units (SU) as the standardized billing abstraction across all network services.

All usage is metered in SU.

Initial cost parameters (subject to node voting):

- **Indexing:** Each new data point = **4,000 SU**.
- **Queries:** Each query = **1,000–2,000 SU**, based on complexity.
- **Node RPC Call:** Each query = **20 SU**

Nodes bid on the cost of SU in \$ST, and jobs are preferentially assigned to nodes with a lower SU cost.

Key benefits:

- **Transparent Pricing:** SU rates are fixed by on-chain governance, ensuring predictability for developers.
- **Flexibility:** Indexers can set their own SU ↔ Token exchange, enabling market-driven efficiency.
- **Extensibility:** New SKUs (e.g., cross-chain queries, AI analytics) can be priced in SU without altering the underlying token model.

5.3 Job Assignment

Job assignment is determined by two factors: stake and SU cost.

A node's capacity allowance is calculated as:

$$Allowance_i = \frac{Stakes_i}{Total\ stakes} Total\ SU\ Consumed\ Last\ Hour$$

A node will stop receiving new jobs, regardless of its SU cost, if its current job count is greater than or equal to its allowance.

For nodes with available capacity and sufficient hardware resources (as configured by the node operator), job assignment favors the one offering the lowest SU cost. Once a job is assigned, the SU cost is locked for that processor for one month.

5.4 Staking Model

- **Sentio Nodes:** Must stake to join; slashing applies for misbehavior.
- **Delegated Staking:** Token holders delegate to Sentio Nodes and share revenue, with commission retained by operators.

5.5 Incentives

- **Sentio Nodes:** Earn SU for indexing and queries.
- **Stakers:** Share Sentio Nodes' revenue.

6. Summary

Sentio is evolving into a decentralized data infrastructure for Web3 through the Sentio Network. Building on a widely adopted, production-grade data and observability platform, Sentio enables developers and protocols to index, process, and analyze blockchain and application telemetry at scale.

The Sentio Network decentralizes the core **storage and processor layers**, moving data persistence and execution from a single service into a network of participants. Processors

execute user-defined data logic off-chain on decentralized nodes, while the blockchain coordinates metadata and verification. This architecture provides transparency, resilience, and shared ownership over critical data pipelines without sacrificing developer experience.

By aligning incentives across node operators, developers, and end-users, the Sentic Network transforms a proven Web3 data platform into an open, network-powered data layer—forming foundational infrastructure for the next generation of decentralized applications.